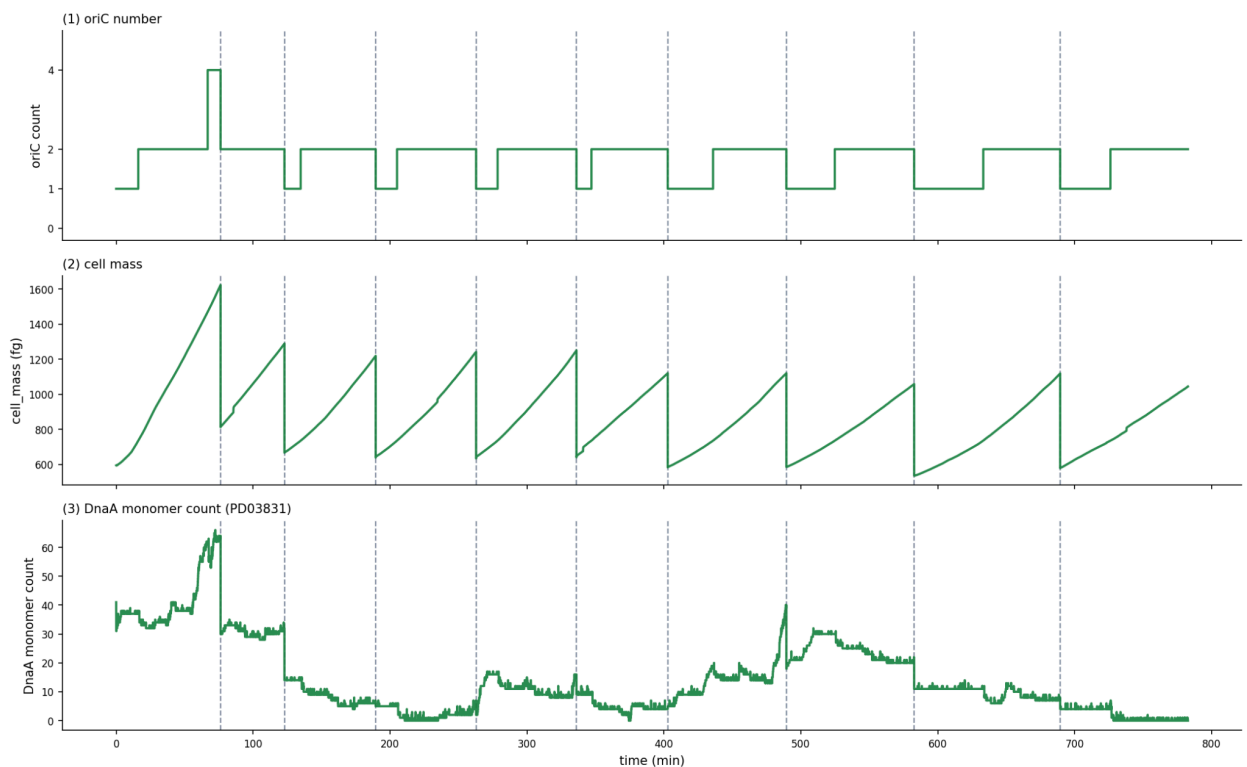


We want to start a new investigation in light of new findings. Please start a brand new investigation without the input files, but bringing in some select context from the current investigation:

- For study dnaa-0: parameter foundations, we do not want glycerol simulations, because v2ecoli is not configured for them. Instead you should use the succinate condition, since succinate already has a saved sim_data state. It is expected to divide in about 70-80 minutes in this condition. Do not use the baseline-stage1-modified parameters for this particular study. These can be used later if needed.
- In continuing dnaa-0, we want to have an initialization task that takes v2ecoli in the succinate condition (run the parca specifying succinate as the growth condition), and run it for 10 generations with single daughters, to check that it converges to a steady state, in which oriC and cell mass display periodic dynamic. After a few generations, the oriC initialization should stabilize 1 oriC, then 2 oriC, but never 4, so the cell cycle will start with one, end with 2 chromosomes with no re-initiations (initiate only once per cell cycle). This will establish that we have steady state succinate dynamics. Save this steady state as the starting point for subsequent simulations in this investigation.
- This is what a steady state simulation (from gen 3 onwards) with periodic dynamics should roughly look like.

Succinate default-fixture — 10-gen parent lineage ($\tau_{\text{target}} = 82$ min, $C = 40$, $D = 20$)



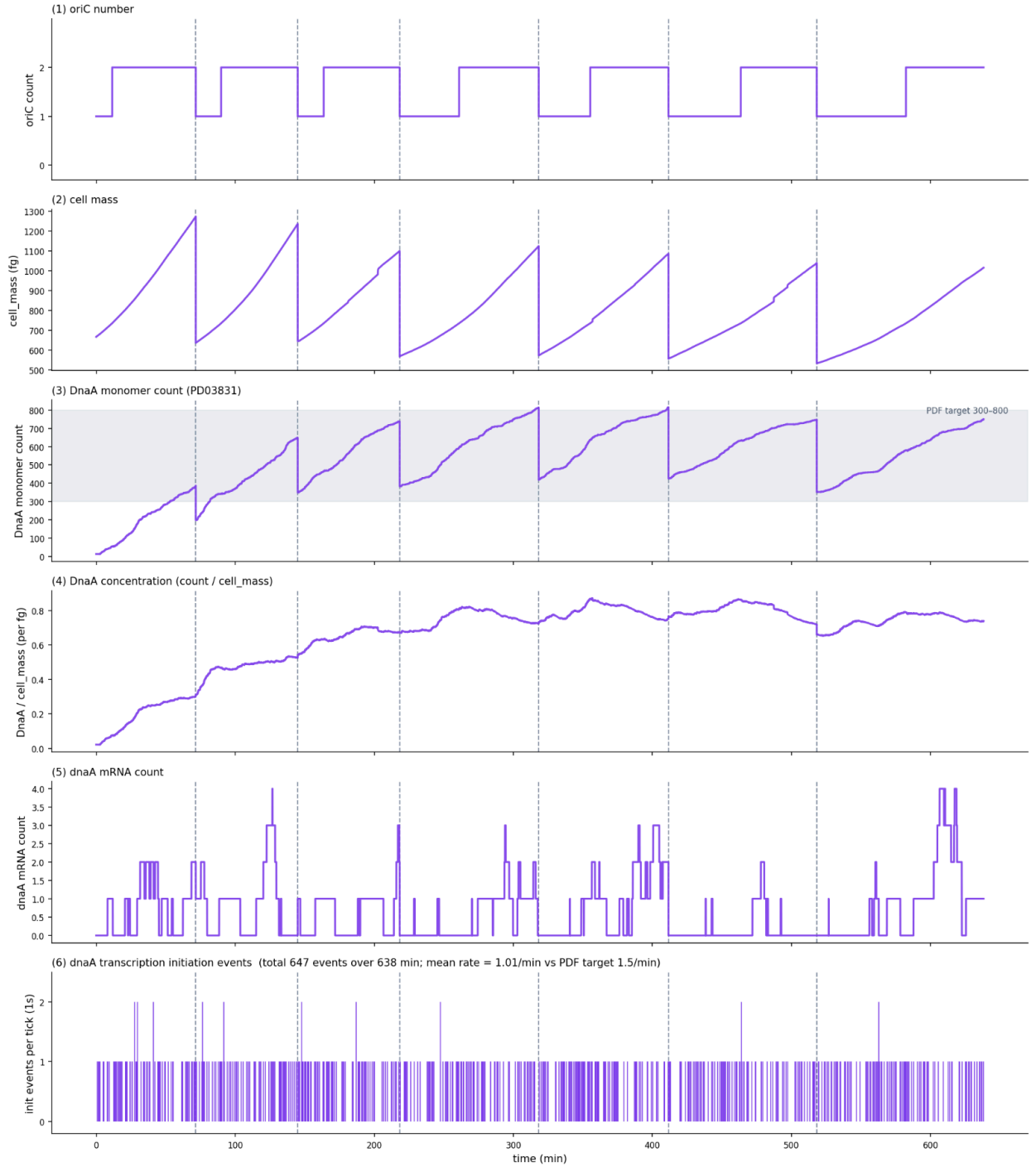
-
- For study dnaa-1, we can move to study 1, which is dnaa expression. For this study, we need to tune the rna synthesis probability to get the dnaa counts within 300-800 counts, and also stable between generations. You can look at this PR for some hints about the working solution for this: <https://github.com/vivarium-collective/v2ecoli/pull/97/changes>

V=2e-3 seed=1 — 7-gen lineage on succinate

CHANGED: `sim_data.genetic_perturbations["TU00259[c]"] = 2e-3` (Mechanism A — runtime overwrite of dnaA's per-promoter init_prob)

UNCHANGED vs baseline: dnaA TE = 0.35 · dnaA mRNA $t_{1/2}$ = 1.9 min · DnaA protein $t_{1/2}$ = 280 min · C = 40 min · D = 20 min

M*/oriC = 734.8 fg · all TF / ppGpp regulation (basal_prob + delta_prob) preserved



- To validate steady behaviour of DnaA expression within expected range, plot counts of DnaA monomer (all DnaA forms - through the monomer counts listener), DnaA concentration (DnaA count/ cell mass), dnaA mRNA count, dnaA mRNA initiation events

- Do not attempt the next studies, until the first two studies pass and are validated.
- We want to completely remove the following studies: dnaa-06
- Do not have RIDA in the glossary or in any of the studies.

Remove prior context and start over fresh